

Weekly Challenge 3

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Given equation:

$$x^2 - 5y^2 = 3$$

Taking mod 5 on both sides,

$$x^2 \bmod 5 - (5y^2) \bmod 5 = 3 \bmod 5$$

Since $5y^2$ is divisible by 5 for all integers y ,

$$5y^2 \bmod 5 = 0$$

Therefore,

$$x^2 \bmod 5 = 3 \bmod 5 \quad (1)$$

Now, x can be written as

$$x = 5k + n, \quad \text{for } n = 0, 1, 2, 3, 4$$

For $n = 0$:

$$x = 5k$$

$$x^2 = 25k^2$$

$$x^2 \bmod 5 = 25k^2 \bmod 5 = 0$$

$$0 \bmod 5 \neq 3 \bmod 5$$

$\therefore$ For $n = 0$, no integer solution.

For $n = 1$:

$$x = 5k + 1$$

$$x^2 = 25k^2 + 10k + 1$$

$$x^2 \bmod 5 = (5(5k^2 + 2k) + 1) \bmod 5 = 1 \bmod 5$$

$$1 \bmod 5 \neq 3 \bmod 5$$

$\therefore$ For $n = 1$, no solution.

For $n = 2$:

$$x = 5k + 2$$

$$x^2 = 25k^2 + 20k + 4$$

$$x^2 \bmod 5 = [5(5k^2 + 4k) + 4] \bmod 5 = 4 \bmod 5$$

$$4 \bmod 5 \neq 3 \bmod 5$$

$\therefore$ For $n = 2$, no solution either.

For $n = 3$:

$$x = 5k + 3$$

$$x^2 = 25k^2 + 30k + 9 = 25k^2 + 30k + 5 + 4 = 5(5k^2 + 6k + 1) + 4$$

$$x^2 \bmod 5 = (5(5k^2 + 6k + 1) + 4) \bmod 5 = 4 \bmod 5$$

$$4 \bmod 5 \neq 3 \bmod 5$$

$\therefore$ For $n = 3$, no solution.

For $n = 4$:

$$x = 5k + 4$$

$$x^2 = 25k^2 + 40k + 16 = 25k^2 + 40k + 15 + 1 = 5(5k^2 + 8k + 3) + 1$$

$$x^2 \bmod 5 = (5(5k^2 + 8k + 3) + 1) \bmod 5 = 1 \bmod 5$$

$$1 \bmod 5 \neq 3 \bmod 5$$

Hence, we can see that for any integer value of x , equation (1) is not satisfied.

$\therefore$ There are no integer solutions (x, y) for $x^2 - 5y^2 = 3$.